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PMC11434556 | Suzuki – Miyaura 偶联 | 采集状态 imported_unchecked

标题	A Ligand-Free Approach towards Coumarin Analogs via Natural Deep Eutectic Solvent-Mediated Suzuki – Miyaura Coupling
作者	Katopodi A., Nikolaou N., Kakokefalou V., Alexandratou E., Matzapetakis M., Zervou M. 等 8 人
出处	Molecules 2024
DOI	10.3390/molecules29184398
原文链接	https://pmc.ncbi.nlm.nih.gov/articles/PMC11434556/
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摘要（原文节选）

A ligand-free approach for the Suzuki-Miyaura cross coupling reaction using Natural Deep Eutectic Solvents (NaDES) towards coumarin analogs is described. A model reaction between the synthetically prepared 3-(4-acetyloxy-phenyl)-6-bromo-4-methyl-coumarin (3b) and phenylboronic acid was performed in five different NaDES as well as in pure glycerol, using two inorganic bases and palladium catalysts. The reaction proceeded smoothly in Choline Chloride/Glycerol (ChCl/Gly) and Betaine/Glycerol (Bet/Gly) NaDES at 90 ° C i ⋯

数据表 · Table 1（10 行，含数值行 5 行）

Tested conditions and yields of the model reaction.

Solvent *	Base	Catalyst (0.025 mol/mol Pd(II)/3b)	Yield (%)
Choline chloride/Glycerol = 1:2	Na2CO3	Pd(OAc)2	85
Betaine/Glycerol = 1:2	Na2CO3	Pd(OAc)2	70
Choline chloride/Glycerol = 1:2	K2CO3	Pd(OAc)2	95
Choline chloride/Glycerol = 1:2	K2CO3	PdCl2	83
Betaine/Glycerol = 1:2	K2CO3	Pd(OAc)2	71
L-proline/Glycerol = 1:2	K2CO3	Pd(OAc)2	traces
L-proline/Glycerol = 1:2	K2CO3	PdCl2	traces
Glycerol	K2CO3	Pd(OAc)2	traces
Fructose/urea/H2O = 1:1.5:1	K2CO3	Pd(OAc)2	No
Glucose/urea/H2O = 1:1.5:1	K2CO3	Pd(OAc)2	No

脚注（逐字）：* The ratios refer to the molar ratio of the components.

正文对应句（逐字）：It can be observed that the reaction proceeded smoothly after 24 h in both Choline chloride/Glycerol (ChCl/Gly) = 1:2 and Betaine/Glycerol (Bet/Gly) = 1:2 NaDES, and the product was obtained in high yields (70 – 95%). In the case of L-proline/Glycerol (Pro/Gly) = 1:2 NaDES as well as in pure glycerol,

数据表 · Table 3（4 行，含数值行 4 行）

Tested conditions, reaction time and yield for the deacetylation reaction.

Reaction Conditions	Solvent	Heating Method	Time	Yield (%)
i	ChCl/Gly = 1:2	Conventional heating (90 ° C)	3 h	79
ii	ChCl/Gly = 1:2	Ultrasound irradiation	15 min	73
iii	Glycerol	Conventional heating (90 ° C)	3 h	72
iv	Glycerol	Ultrasound irradiation	6 min	92

正文对应句（逐字）：It can be observed that the reaction proceeded smoothly after 24 h in both Choline chloride/Glycerol (ChCl/Gly) = 1:2 and Betaine/Glycerol (Bet/Gly) = 1:2 NaDES, and the product was obtained in high yields (70 – 95%). In the case of L-proline/Glycerol (Pro/Gly) = 1:2 NaDES as well as in pure glycerol,

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